FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 4: HOMOMORPHISMS, RINGS, FIELDS AND INTEGRAL DOMAINS

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

1 Define the kernel of a group homomorphism $\phi : G \to G'$. **(1 mark)**

2 State whether true or false: in every ring with unity, the element 1 is a unit. **(1 mark)**

3 Find the characteristic of the ring $\mathbb{Z}_6 \times \mathbb{Z}_{10}$. **(1 mark)**

4 Is $\{0, 1\}$ with addition and multiplication modulo 2 a field? Answer yes or no. **(1 mark)**

Section B: 2 marks each, answer all

5 Let $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_4$ be the group homomorphism with $\phi(1) = 1$. Find $\operatorname{Ker} \phi$. **(2 marks)**

6 State the conditions a map $\phi : R \to R'$ must satisfy to be a ring homomorphism. **(2 marks)**

7 Find all units of the ring $\mathbb{Z}_{10}$. **(2 marks)**

8 Show that the ring $\mathbb{Z}_2$ is an integral domain. **(2 marks)**

9 Let $\phi : \mathbb{Z}_{10} \to \mathbb{Z}_5$ be $\phi(x) = x \bmod 5$. Is ϕ a group homomorphism? Justify. **(2 marks)**

Section C: 4 marks each, answer any two

10 Let $\phi : G \to G'$ be a group homomorphism. Show that $\phi(x) = \phi(y)$ if and only if $\phi(xy^{-1}) = e'$. **(4 marks)**

11 Let R be a ring with unity and $\phi : R \to R'$ a ring homomorphism onto R'. Show that $\phi(1)$ is unity of R'. **(4 marks)**

12 Show that the set $\{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{R}$. **(4 marks)**

13 Let $\phi : G \to G'$ be a homomorphism. Show that $\operatorname{Ker} \phi$ is a normal subgroup of G. **(4 marks)**

14 Show that a finite commutative ring with unity and no zero divisors is a field. **(4 marks)**

Section D: 10 marks each, answer any one

15 State and prove the First Isomorphism Theorem for groups, and use it to show that $\mathbb{Z}_{12}/\langle 4 \rangle \cong \mathbb{Z}_4$ by listing the map. **(10 marks)**

16 Let p be a prime. Show that $\mathbb{Z}_{p^2}$ has $p^2 - p$ units, and that the nonzero elements of $\mathbb{Z}_{p^2}$ that are not units are exactly the multiples of p. **(10 marks)**

MODULE 4: MODEL ANSWERS

Key steps for checking your work

Section A.

1. $\mathbf{Ker}\,\phi = \phi^{-1}[\{e'\}] = \{x \in G \mid \phi(x) = e'\}$ (Definition 8.6).
2. True. $1 \cdot 1 = 1$, so 1 has a multiplicative inverse, namely itself (Definition 22.16).
3. 30. The characteristic of $\mathbb{Z}_6 \times \mathbb{Z}_{10}$ is the least n with $n \cdot 1 = 0$, which is $\mathbf{lcm}(6, 10) = 30$ (Definition 23.12; Theorem 23.14).
4. Yes. $\{0, 1\}$ is $\mathbb{Z}_2$, a field (Corollary 23.5).

Section B.

5. $\phi(x) = x \bmod 4$ on $\mathbb{Z}_{12}$ sends x to the class of x in $\mathbb{Z}_4$; the kernel is $\{0, 4, 8\} = \langle 4 \rangle$ (Definition 8.6).
6. $\phi(a + b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$ for all a, b (Definition 22.9).
7. $\{1, 3, 7, 9\}$, the numbers relatively prime to 10 (Theorem 23.3).
8. $\mathbb{Z}_2$ is commutative with unity $1 \neq 0$ and has no divisors of 0: the only nonzero element is 1, and $1 \cdot 1 = 1 \neq 0$ (Definition 23.7).
9. Yes. Since $5 \mid 10$, the value $x \bmod 5$ does not depend on the representative of x in $\mathbb{Z}_{10}$, so ϕ is well defined. And $\phi(x + y) = \phi(x) + \phi(y)$ in $\mathbb{Z}_5$ (Definition 8.1).

Section C.

10. $\phi(xy^{-1}) = \phi(x)\phi(y)^{-1}$ by Theorem 8.5 (2). This equals e' exactly when $\phi(x)\phi(y)^{-1} = e'$, that is when $\phi(x) = \phi(y)$.
11. For $r' \in R'$, $r' = \phi(r)$ for some r, and $\phi(1)\phi(r) = \phi(r) = \phi(r)\phi(1)$. So $\phi(1)$ is unity of R' (Definition 22.9, onto).
12. $0 \in$ the set; the difference and the product of two elements are again of this form, since $(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$ (Section 22, subring test).
13. $\mathbf{Ker}\,\phi$ is a subgroup (Theorem 8.5 (4)). For $g \in G$, $k \in \mathbf{Ker}\,\phi$: $\phi(gkg^{-1}) = \phi(g)e'\phi(g)^{-1} = e'$. So it is normal (Exercise 35 of Section 8).
14. Let $a \neq 0$. Multiplication by a is one-to-one (cancellation, Theorem 23.6) and so onto (finite set). So some b has $ab = 1$ (Theorem 23.11), and the ring is a field.

Section D.

15. Let $\phi : \mathbb{Z}_{12} \to \mathbb{Z}_4$ be reduction mod 4; ϕ is a homomorphism with kernel $\langle 4 \rangle$ (Theorem 16.2 and Example 16.7). The map $\mu(x + \langle 4 \rangle) = \phi(x)$ is an isomorphism $\mathbb{Z}_{12}/\langle 4 \rangle \to \mathbb{Z}_4$ (First Isomorphism Theorem).
16. $\mathbb{Z}_{p^2}$ has p^2 elements. By Theorem 23.3 the units are the classes of numbers coprime to p^2, which are the $p^2 - p$ numbers not divisible by p. The non-units are the multiples of p, which are p numbers (including 0).

MODULE 4: SYLLABUS MAPPING

Topics the past papers do not reach, and the model questions that cover them

Topic	**Book section (8th ed.)**	**Model questions**
Kernel and the definition of a group homomorphism	8.1, 8.6	1, 5, 10
Ring homomorphisms	22.9	6, 9, 11
Kernel is a normal subgroup	8.6, 13.18 (normal subgroups; check against Module V)	13
Fundamental (first isomorphism) theorem	16.2	15
Subrings and units	22.2, 22.16, 23.3	7, 12
Integral domains, fields and cancellation	23.6, 23.7, 23.11	8, 14
Characteristic	23.12 to 23.14	3, 4
Units and divisors of zero in Z_n	23.3, 23.5	16